

Technical note: searching the ASTRA quantile data vectors for the largest Fisher gain over the full-sample 2PCF

ASTRA clustering project

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Abstract

Using the complete grid of full-box runs on disk (nine AbacusSummit cosmologies $\times$ 50 HOD draws), we ask which ASTRA environment-split statistic, or which pair of statistics, extracts the most cosmological Fisher information beyond the plain full-sample auto-correlation. We sweep all 16 single quantile vectors and all 120 pairs, in two framings (standalone and added to the full auto), scoring each by the HOD-marginalised Fisher figure of merit. The naive ranking is dominated by the *random*-quantile legs, with apparent gains up to $55\times$; we show this is an artefact of unmodelled derivative noise in those noise-limited legs (all of the global top 20 carry a random leg), not real information. Among the trustworthy *data*-leg vectors the gains are real: adding a single full $\times$ data-quantile cross buys $\sim 5\times$ in the three-parameter FoM, and a greedy forward selection that spans the four density quantiles reaches a five-stem vector (`full+xdQ1+dQ1+dQ2+xdQ3+dQ4`) with $\text{FoM}_3 = 43\times$ — marginalised errors tighter than the standard 2PCF by $3.4\times$ (ω_b), $4.1\times$ (ω_c), $3.6\times$ (n_s) and $2.8\times$ (σ_8), with clear diminishing returns past four stems. We document the method, the artefact diagnosis, the honest ranking, and the greedy chain.

1 Question and data

ASTRA splits a galaxy sample into four environment quantiles (Q1 underdense $\rightarrow$ Q4 overdense) and measures, per box, the multipoles $\xi_{0,2}(s)$ of: the full-sample auto-correlation (`tpcf_full_data`, our baseline), the four data-quantile autos (`dQ1..4`), the four random-quantile autos (`rQ1..4`), the four full $\times$ data-quantile crosses (`xdQ1..4`), and the four full $\times$ random-quantile crosses (`xrQ1..4`). That is 17 stems $\times \{\ell = 0, 2\} \times 15$ bins per run.

The signal data are the $9 \times 50 = 450$ phase-matched full-box runs (`data/fullbox/`); cosmology derivatives come from the global linear response fit (`derivative_global_*`, condition number 1.34 on the full grid). The covariance of one $500 h^{-1}$ Mpc subbox is estimated from the mean-subtracted subboxes pooled across all nine cosmologies, $9 \times 64 = 576$ fluctuation samples, and scaled to the full $2000 h^{-1}$ Mpc box volume ($C/64$). The HOD nuisance covariance C_{HOD} is the empirical scatter of the 50 same-phase c000 draws.

2 Method

The analysis is implemented in `scripts/fisher_vector_search.py`, which reuses the assembly and Fisher machinery of `scripts/fisher_joint.py`. The steps are:

1. **Candidate vectors.** Each candidate is one or two ASTRA stems, taken at *native* 15-bin resolution in mono+quad ($\ell = 0, 2$). The 576-sample pooled covariance keeps the Hartlap

Table 1: Raw top of the additive ranking (full auto + stem(s)). A star marks a candidate that contains a random leg (rQ or xrQ). *Every* entry in the global top eight, and all 20 of the global top 20, carry a random leg, despite random legs being the weakest tracers in every earlier single-parameter test.

	vector	FoM ₃ gain	FoM ₄ gain
singles	xrQ4 [*]	12.2	13.5
	rQ4 [*]	8.6	9.9
	xrQ1 [*]	7.8	8.6
pairs	xrQ1+xrQ4 [*]	55.8	64.6
	xdQ1+xrQ4 [*]	44.6	59.0
	xdQ4+xrQ4 [*]	43.4	60.3

factor benign even at $n_b = 60$ (pair) or 90 (pair added to the full auto), so no rebinning is needed. We evaluate the 16 singles and all $\binom{16}{2} = 120$ pairs.

2. **Two framings.** *Standalone* scores the candidate by itself against the full auto (“is it better than the 2PCF?”). *Additive* scores the full auto with the candidate appended (“what does ASTRA add to the 2PCF?”). The additive framing is the physical measure of marginal information and is our headline.
3. **Fisher and marginalisation.** For each candidate we build the 16×16 joint Fisher over the four cosmological parameters $\{\omega_b, \omega_c, n_s, \sigma_8\}$ and the twelve yuan23 HOD parameters, $F = D C^{-1} D^T$ with Hartlap-corrected precision, add a Gaussian yuan23 prior on the HOD block, and read off the HOD-marginalised 4×4 cosmology covariance $\text{Cov} = (F + F_{\text{prior}})_{[\text{cosmo}]}^{-1}$ (route a).
4. **Figure of merit.** We reduce Cov to

$$\text{FoM}_4 = [\det \text{Cov}_4]^{-1/2}, \quad \text{FoM}_3 = [\det \text{Cov}_3]^{-1/2}, \quad (1)$$

where Cov₃ drops σ_8 (the upper-left 3×3 block, already marginalised over σ_8). The *gain* of a candidate is $\text{FoM}/\text{FoM}_{\text{full auto}}$. We headline FoM₃ because the σ_8 derivative is unreliable (Sec. 3), so a four-parameter FoM inherits that unreliability.

3 The random-leg trap

Table 1 and the bright rQ4/xrQ4/xrQ1 band in Fig. 2 show that the naive sweep is dominated by random legs, with implausible gains (FoM₃ up to 55 $\times$, i.e. a factor ~ 4 on every parameter from one box). This contradicts every prior result, in which the underdense random leg was the *weakest* vector and carried the most derivative noise. The resolution is that these gains are an artefact:

- The random quantiles — especially their quadrupoles — are *noise-limited*: their per-bin ASTRA-random scatter is comparable to the HOD-induced signal (measured separately: rQ2/3 $\ell=2$ noise/ signal $\approx 0.7\text{--}0.8$). Their global-fit derivatives therefore contain large noise components.

Table 2: Data-leg candidates only (no random leg). Additive framing, HOD-marginalised. Baseline full-auto errors: $\sigma(\omega_b) = 5.7 \times 10^{-4}$, $\sigma(\omega_c) = 1.7 \times 10^{-3}$, $\sigma(n_s) = 1.3 \times 10^{-2}$, $\sigma(\sigma_8) = 1.4 \times 10^{-2}$.

	vector	FoM ₃ gain	$\sigma(\omega_b)$	$\sigma(\omega_c)$	$\sigma(n_s)$
singles	xdQ1	5.1	3.3×10^{-4}	8.4×10^{-4}	8.3×10^{-3}
	xdQ4	5.0	3.2×10^{-4}	8.4×10^{-4}	8.5×10^{-3}
	dQ4	4.1	3.3×10^{-4}	9.8×10^{-4}	8.4×10^{-3}
	dQ1	3.9	4.2×10^{-4}	1.1×10^{-3}	7.7×10^{-3}
pairs	dQ1+xdQ1	14.0	2.8×10^{-4}	—	6.0×10^{-3}
	dQ2+xdQ1	12.8	2.5×10^{-4}	—	4.9×10^{-3}
	xdQ3+xdQ4	12.2	2.6×10^{-4}	—	5.0×10^{-3}

- A Fisher matrix treats the derivative D as exact. Noise in D that is not represented in the data covariance C inflates $D C^{-1} D^T$: the Fisher reads derivative noise as constraining power, and a noisy derivative pointing in a direction nearly orthogonal to the full auto spuriously “breaks” degeneracies.
- The signature is diagnostic: a random leg is barely better than the full auto on its own (**xrQ4** standalone FoM₃ = 1.6) yet explodes when appended (12×), and *all* 20 top-20 vectors carry a random leg. Real information would not concentrate so completely in the lowest-S/N legs.

We therefore flag every random-leg candidate and base the physical conclusions on the data legs only. Removing the random-leg artefact is exactly the role of propagating the emulator derivative variance (future work); it would deflate these spurious gains.

4 The honest, data-leg ranking

Restricting to data-leg candidates (Table 2, Fig. 1) gives a clean and physically sensible picture:

- **Singles.** The full×data-quantile crosses at the extremes lead — **xdQ1** (void) and **xdQ4** (knot), FoM₃ gain ≈ 5 — followed by the extreme data autos **dQ4**, **dQ1** (≈ 4). The middle quantiles (Q2, Q3) are weaker. Each of these improves ω_b and n_s by ~ 1.6 – $1.7\times$ over the full auto. Note that on its own each vector is *worse* than the full auto (standalone FoM₃ < 1.1); the gain is entirely degeneracy-breaking when added — the ASTRA thesis.
- **Pairs.** The best clean pair is **dQ1+xdQ1** (void data auto + void data cross), FoM₃ gain 14, i.e. a factor ~ 2.0 on ω_b and ~ 2.2 on n_s over the full auto. Combinations of an extreme data auto with an extreme data cross dominate; pairs of two middle quantiles, or two stems from the same family, add little (redundant).
- **σ_8 is untouched.** No vector improves σ_8 appreciably ($1.4 \times 10^{-2} \rightarrow \sim 1.0 \times 10^{-2}$), consistent with σ_8 being amplitude-degenerate with the HOD and with its derivative being unreliable. This is why FoM₃ (which drops σ_8) is the trustworthy score; Fig. 3 confirms the per-parameter picture.

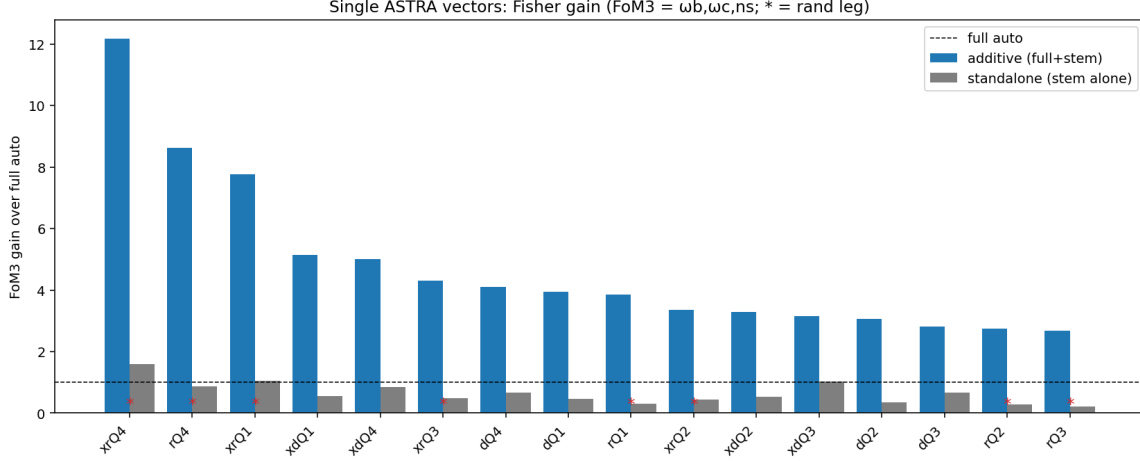


Figure 1: Single-vector FoM₃ gain over the full auto, additive (blue) and standalone (grey), ranked. Red stars mark random-leg vectors, whose additive gains are inflated by derivative noise (Sec. 3). Every vector is at or below the full-auto line on its own; the gain is degeneracy-breaking, realised only when appended.

Table 3: Greedy chain over the data-leg stems. “marg.” is the multiplicative gain in FoM₃ from that one step. σ values are the HOD-marginalised errors of the cumulative vector. The chain visits all four density quantiles by step five.

step	add	meaning	FoM ₃	marg.	$\sigma(\omega_b)$	$\sigma(n_s)$	$\sigma(\sigma_8)$
0	—	full auto	1.0	—	5.7×10^{-4}	1.3×10^{-2}	1.4×10^{-2}
1	xdQ1	void cross	5.1	5.1	3.3×10^{-4}	8.3×10^{-3}	1.0×10^{-2}
2	dQ1	void auto	14.0	2.7	2.8×10^{-4}	6.0×10^{-3}	7.5×10^{-3}
3	dQ2	low–mid auto	24.7	1.8	2.3×10^{-4}	4.3×10^{-3}	6.8×10^{-3}
4	xdQ3	mid–high cross	32.5	1.3	2.1×10^{-4}	3.8×10^{-3}	5.4×10^{-3}
5	dQ4	knot auto	43.3	1.3	1.7×10^{-4}	3.6×10^{-3}	5.0×10^{-3}

5 Greedy forward selection: how many stems, and which?

The single- and pair-level results raise the obvious question: if one extreme data cross already buys $\sim 5\times$ and the void auto+cross pair buys $14\times$, how far does it go? We answer it with a *greedy forward selection* restricted to the eight data-leg stems (`scripts/fisher_greedy_chain.py`). Starting from the full auto, at each step we append the single data-leg stem that most increases FoM₃, keep it, and repeat, up to five added stems. Greedy selection is not guaranteed to find the globally optimal subset, but it is the natural way to read off the *marginal* value of each successive statistic and to expose redundancy.

The chain is `full+xdQ1+dQ1+dQ2+xdQ3+dQ4` (Table 3, Fig. 4). Three things stand out.

- **Strong diminishing returns.** The per-step gain falls $5.1 \rightarrow 2.7 \rightarrow 1.8 \rightarrow 1.3 \rightarrow 1.3$. The first two stems carry most of the information (FoM₃ reaches 14 of the eventual 43 by step two); beyond step four each new statistic adds only $\sim 30\%$, while the bin count and Hartlap/redundancy cost keep rising. A four-stem vector is the practical sweet spot.
- **Gains come from spanning environments, not repetition.** Greedy does not pile up void

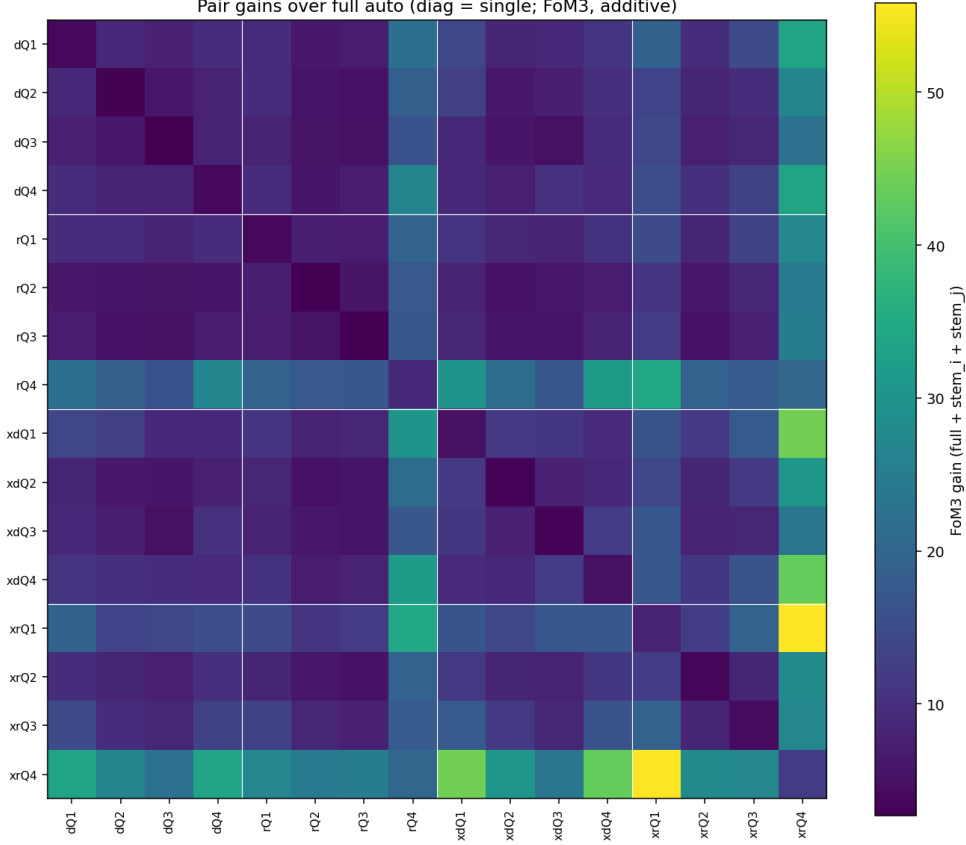


Figure 2: Additive FoM₃ gain for every pair, full auto + stem_i + stem_j (diagonal = single). White lines separate the four families (dQ, rQ, xdQ, xrQ). The bright band on the rQ4/xrQ4/xrQ1 rows and columns is the random-leg artefact; the data-leg quadrants (dQ, xdQ) show the modest, trustworthy gains.

statistics; after the void auto+cross it reaches for a *different* density bin (dQ2), then mid-high (xdQ3), then the knot (dQ4), touching all four quantiles by step five. Different environments respond to cosmology and to the HOD in different ways, so each new environment breaks a *complementary* degeneracy; a second statistic from the same environment would be largely redundant.

- **σ_8 moves only in combination.** No single vector improved σ_8 (Sec. 4), but the cumulative five-stem vector tightens it $2.8 \times (1.4 \times 10^{-2} \rightarrow 5.0 \times 10^{-3})$. The multi-environment combination partially lifts the amplitude degeneracy that defeats any one statistic — though σ_8 remains the least trustworthy entry because its derivative is unreliable, and FoM₃ (which excludes it) stays our headline score.

The five-stem vector versus the 2PCF

Figure 5 is the head-to-head Fisher comparison of the standard full-sample 2PCF (tpcf_full_data, mono+quad) and the five-stem ASTRA vector, both HOD-marginalised, generated by scripts/fisher.compare.fu. The five-stem 68/95% contours sit nested well inside the full-auto contours on every panel, with the marginalised errors tighter by

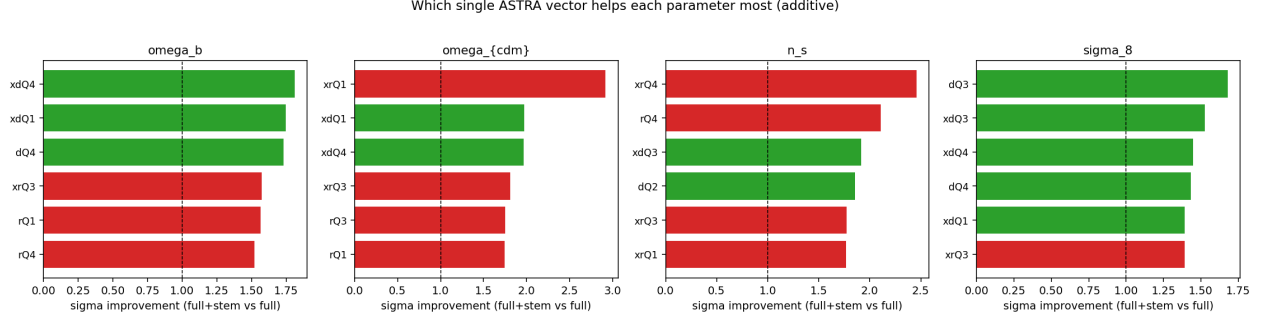


Figure 3: For each cosmological parameter, the single ASTRA vectors that most improve its marginalised error when added to the full auto. Green bars are data-leg vectors (trustworthy); red bars are random-leg vectors (artefact).

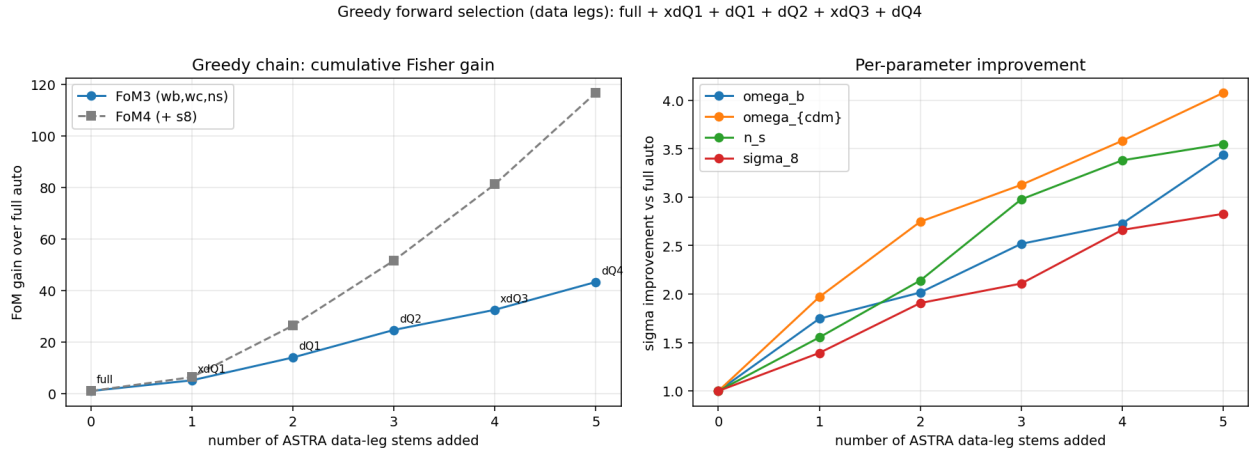


Figure 4: Greedy forward selection over the data legs. Left: cumulative FoM₃ (and FoM₄) gain over the full auto versus the number of ASTRA stems added — the curve flattens after ~ 4 stems. Right: per-parameter σ improvement over the full auto. Labels mark the stem chosen at each step.

$$\sigma(\omega_b) \ 3.4\times, \quad \sigma(\omega_c) \ 4.1\times, \quad \sigma(n_s) \ 3.6\times, \quad \sigma(\sigma_8) \ 2.8\times, \quad (2)$$

i.e. FoM₃ = 43 $\times$. Crucially the blue ellipses are not merely smaller but *rotated* relative to grey: the environment splits change the orientation of the parameter degeneracies, which is the geometric signature of genuinely new information rather than a uniform rescaling. The five-stem 95% region is inside the full-auto 68% region for most parameter pairs — a qualitatively different constraint from a single $2\ h^{-1}$ Mpc box, by reanalysis alone.

6 Conclusions

1. The systematic sweep of all 16 singles and 120 pairs is dominated, at face value, by random-leg vectors with gains up to 55 $\times$. This is a **derivative-noise artefact**, not physics: the noise-limited random legs have noisy global-fit derivatives that the Fisher misreads as information. Any data-vector ranking on this grid must exclude or noise-correct the random legs.
2. Among the **trustworthy data legs**, the full $\times$ data-quantile crosses at the density extremes

(**xdQ1**, **xdQ4**) are the best single additions (FoM₃ gain ≈ 5), and the best clean pair is the void data auto with the void data cross, **dQ1+xdQ1** (FoM₃ gain 14, $\sim 2.4\times$ per parameter on ω_b, ω_c, n_s). The gains are pure degeneracy breaking: each vector is worse than the full auto alone.

3. A **greedy chain** over the data legs (**full+xdQ1+dQ1+dQ2+xdQ3+dQ4**) reaches FoM₃ = $43\times$ at five stems, tightening the standard-2PCF errors by $3.4\text{--}4.1\times$ on ω_b, ω_c, n_s and $2.8\times$ on σ_8 (Fig. 5). Returns diminish sharply ($5.1 \rightarrow 2.7 \rightarrow 1.8 \rightarrow 1.3$ per step), so ~ 4 stems is the practical optimum, and the gains come from *spanning* the four environments rather than repeating one.
4. Single vectors do not improve σ_8 (HOD-amplitude degeneracy, unreliable derivative), but the multi-environment combination does so modestly; FoM₃ excluding σ_8 remains the trustworthy score.
5. **Next steps.** (i) Propagate the GP emulator derivative variance (calibrated with the $\sim 1.25\times$ inflation measured separately) so the random-leg artefact is deflated and the random legs can be ranked honestly; (ii) use the higher-iteration runs to denoise the random-quantile derivatives directly; (iii) revisit the four-parameter FoM once a trustworthy σ_8 derivative is available.

All numbers are reproduced by `scripts/fisher_vector_search.py` (the full sweep; ranked table in `plots/vector_search/vector_search_results.csv`), `scripts/fisher_greedy_chain.py` (the greedy chain), and `scripts/fisher_compare_full_vs_5stem.py` (the corner comparison).

Fisher 68/95%: full-auto 2PCF vs 5-stem ASTRA (full + xdQ1 + dQ1 + dQ2 + xdQ3 + dQ4), HOD-marginalised

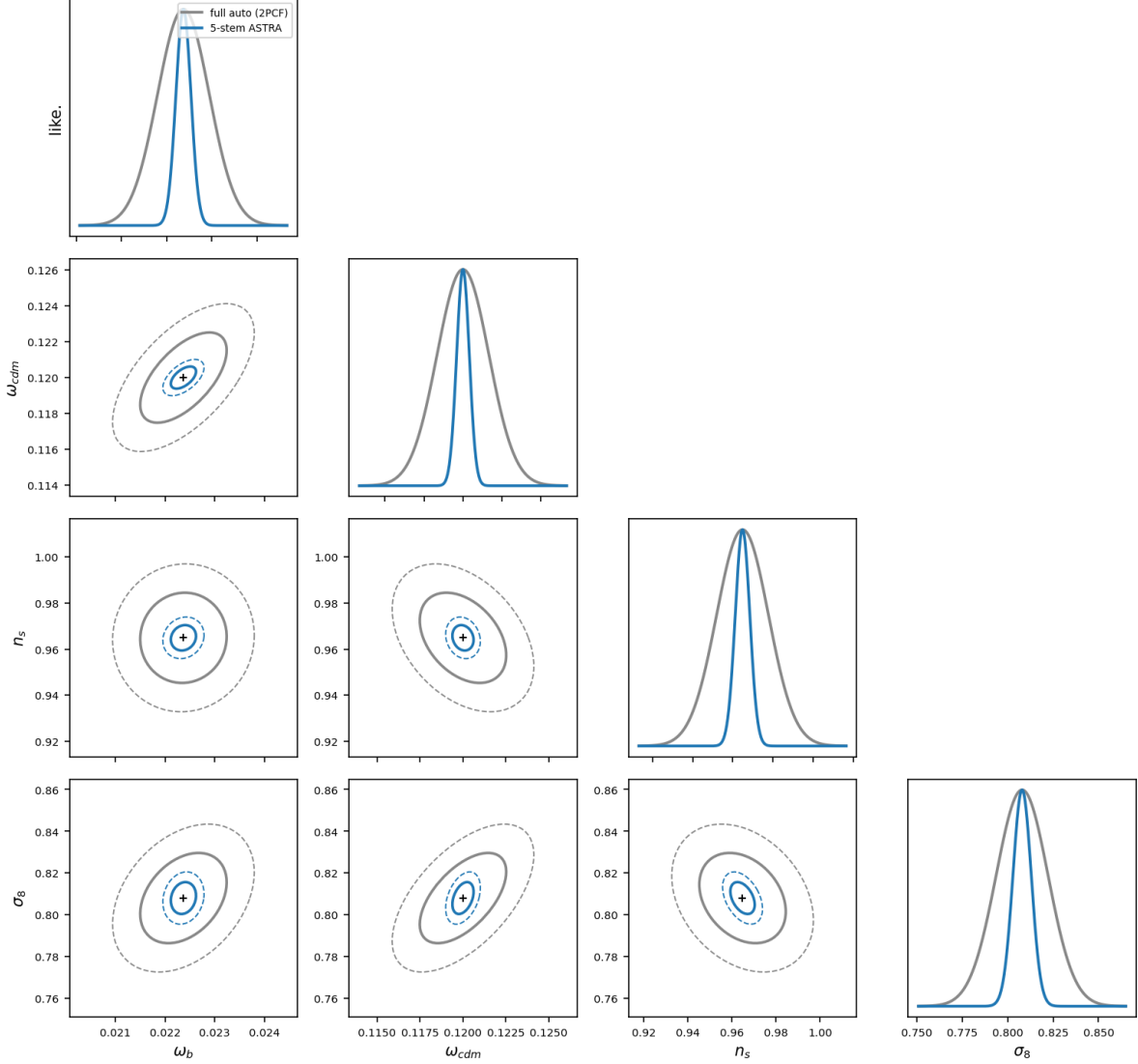


Figure 5: HOD-marginalised Fisher constraints: full-auto 2PCF (grey) versus the five-stem ASTRA vector **full+xdQ1+dQ1+dQ2+xdQ3+dQ4** (blue), 68% (solid) and 95% (dashed). The blue contours are both smaller and rotated — ASTRA shrinks and re-orientes the cosmology degeneracies.